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FEUP FACULDADE DE ENGENHARIA UNIVERSIDADE DO PORTO
DEI DEPARTAMENTO DE ENGENHARIA INFORMÁTICA

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FC FACULDADE DE CIÊNCIAS UNIVERSIDADE DO PORTO
[dcc] DEPARTAMENTO DE CIÊNCIA DE COMPUTADORES FACULDADE DE CIÊNCIAS DA UNIVERSIDADE DO PORTO

Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

# Extra 2 - Proofs in Propositional Logic

All deductions must use the DN natural deduction system and the Fitch notation given in class.

**Notes for obtaining a deduction from a formula $\varphi$**

- If the formula $\varphi$ is an implication $\psi \rightarrow \theta$, assume $\psi$ and deduce $\theta$; then apply the rule of introduction of implication.
- If the formula $\varphi$ is a negation $\neg\psi$, assume $\psi$ and deduce $\mathsf{F}$; then apply the rule of introduction of negation.
- By contradiction: assume $\neg\varphi$ and deduct $\mathsf{F}$. In case there is a negation $\neg\psi$ in the premises, one can deduce $\psi$ to get $\mathsf{F}$.
- Suppose one of the premises is $\varphi \vee \psi$ and one wants to deduce $\gamma$. We can: assume $\varphi$ and deduce $\gamma$, assume $\psi$ and deduce $\gamma$; then apply the rule of elimination of the disjunction.
- Suppose an assumption is $\neg\varphi$ and we want to get $\mathbf{F}$: we can try to get $\varphi$ and apply the intro rule of $\mathbf{F}$.
- Suppose we have an implication $\phi \rightarrow \theta$. We can then try to deduct $\phi$ and apply the rule of elimination of implication (modus ponens) to get $\theta$.

Note that each step of the deduction must be obtained by applying a rule or be a repetition of a formula already deduced or a premise. Thus each deduction step must either be an assumption or result from the application of a rule to formulas already deduced, so "semantic equivalences" cannot be used.

**Note:** In case you use derived rules, you should prove then separately.

**2.1.** Some of the following arguments are valid and some are not. Use symbols to write the logical form of each one. If the argument is valid, identify the inference rule.

(a) If Hugo solves the exercises correctly, he will travel to the Azores. Hugo is going to travel to the Azores. So, Hugo correctly solved the exercises.

(b) If $x$ is greater than 2, then its square is greater than 4. The number $x$ is not greater than 2. Then the square of $x$ is not greater than 4.

(c) If I go to the movies, I don't solve the exercises. If I don't solve the exercises I will fail math. So if I go to the movies I'm going to fail math.

**2.2.** Use the inference rules studied in lectures to complete the following arguments so that they are valid.

(a) If math is hard, then Santa Claus exists. Santa Claus doesn't exists. Therefore...

(b) If math is hard, then Santa Claus exists. Math is hard. Therefore...

**2.3.** Show that:

(a) $\delta \to (\varphi \to \psi), \neg\psi, \delta \vdash \neg\varphi$

(b) $\vdash (\varphi \vee \psi) \to (\psi \vee \varphi)$

(c) $\vdash (\delta \to (\varphi \to \psi)) \to (\varphi \to (\delta \to \psi))$

(d) $\vdash \delta \to (\delta \vee \varphi)$

(e) $\vdash ((\varphi \to \delta) \to \delta) \to ((\delta \to \varphi) \to \varphi)$

(f) $\delta, \varphi \to \psi \vdash \varphi \to (\delta \to \psi)$

(g) $\delta, \neg\varphi \vdash \neg(\delta \to \varphi)$

**2.4.** Show, without using the completeness of DN, the following deductive equivalences. Note that you have to show that the second formula is deduced from the first formula and vice versa.

(a) $\neg\delta_1 \wedge \neg\delta_2 \vdash \delta_1 \to \delta_2$

(b) $\delta_1 \wedge \neg\delta_2 \vdash \neg(\delta_1 \wedge \delta_2)$

(c) $(\psi \vee \varphi) \vee \delta \dashv\vdash \psi \vee (\varphi \vee \delta)$

(d) $\delta \to \varphi \dashv\vdash (\delta \to (\delta \wedge \varphi)) \wedge ((\delta \wedge \varphi) \to \delta)$

(e) $\delta \to \varphi \dashv\vdash (\varphi \to (\delta \vee \varphi)) \wedge ((\delta \vee \varphi) \to \varphi)$

(f) $(\delta \to \varphi) \wedge (\varphi \to \delta) \dashv\vdash (\delta \vee \varphi) \to (\delta \wedge \varphi)$

(g) $\varphi \wedge (\delta \vee \psi) \dashv\vdash (\varphi \wedge \delta) \vee (\varphi \wedge \psi)$

**2.5.** Show, without using the completeness of DN, that for every set of formulas in propositional logic $\Sigma$ and formula $\delta$, you have:

(a) If $\Sigma \vdash \neg\delta$ then $\Sigma \cup \{\delta\}$ is inconsistent.

(b) $\Sigma \models \delta$ if and only if $\Sigma \cup \{\neg\delta\}$ is unsatisfiable.

(c) $\Sigma \cup \{\neg\delta\} \vdash \mathbf{F}$ if and only if $\Sigma \vdash \delta$

**Natural Deduction (DN) for propositional logic (Fitch notation)**

| | **Introduction** | **Elimination** |
|---|---|---|
| $\wedge$ | $\varphi$<br>$\vdots$<br>$\psi$<br>$\vdots$<br>$\varphi \wedge \psi$ $\wedge$I | $\vdots$<br>$\varphi \wedge \psi$<br>$\vdots$<br>$\varphi$ $\wedge$E  $\vdots$<br>$\varphi \wedge \psi$<br>$\vdots$<br>$\psi$ $\wedge$E |
| $\vee$ | $\vdots$<br>$\varphi$<br>$\vdots$<br>$\varphi \vee \psi$ $\vee$I  $\vdots$<br>$\varphi$<br>$\vdots$<br>$\psi \vee \varphi$ $\vee$I | $\varphi \vee \psi$<br>$\vdots$<br>$\varphi$ / $\vdots$ / $\gamma$<br>$\psi$ / $\vdots$ / $\gamma$<br>$\gamma$ $\vee$E |
| $\neg$ | $\varphi$ / $\vdots$ / **F**<br>$\neg\varphi$ $\neg$I | $\vdots$<br>$\neg\neg\varphi$<br>$\vdots$<br>$\varphi$ $\neg$E |
| **F** | $\varphi$<br>$\vdots$<br>$\neg\varphi$<br>$\vdots$<br>**F FI** | $\vdots$<br>**F**<br>$\vdots$<br>$\varphi$ **FE** |
| $\rightarrow$ | $\varphi$ / $\vdots$ / $\psi$<br>$\varphi \rightarrow \psi$ $\rightarrow$I | $\varphi$<br>$\vdots$<br>$\varphi \rightarrow \psi$<br>$\vdots$<br>$\psi$ $\rightarrow$E |